OBJECTS THAT BECOME ZERO IN THE LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1788-1810, following the statement at 1765-1786.

Editorial note. No omissions: (6) is explicitly marked dual to (5), following the established dual-mirror counting practice.

Allowed standard prerequisites (editorial).

- Kernel localization in Homology. Homology, Lemma homology-lemma-kernel-localization (cross-chapter, cited at ~1788). Gives the equivalence of (1), (2), (3).
- Saturation and inversion. Categories, Lemma categories-lemma-what-gets-inverted (cross-chapter, cited at ~1802). Under saturation, morphisms that become isomorphisms lie in S, giving (1) => (5).
- Split triangles. lemma-split (same chapter, cited at ~1793). Provides the split distinguished triangle used for (2) => (4).

Source reference. Original reference 05Q9: focused-seeds/geometry/S9-der-kernel-localization/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05QF: focused-seeds/geometry/S9-der-kernel-localization/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QT: sources/stacks-derived/pinned-original-source.tex, source lines 917–917. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let S be a multiplicative system compatible with the triangulated structure. Let Z be an object of $\mathcal{D}$. The following are equivalent*

(1) *$Q(Z) = 0$ in $S^{-1}\mathcal{D}$,*
(2) *there exists $Z' \in \mathrm{Ob}(\mathcal{D})$ such that $0 : Z \to Z'$ is an element of S,*
(3) *there exists $Z' \in \mathrm{Ob}(\mathcal{D})$ such that $0 : Z' \to Z$ is an element of S, and*
(4) *there exists an object Z' and a distinguished triangle $(X, Y, Z \oplus Z', f, g, h)$ such that $f \in S$.*

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If S is saturated, then these are also equivalent to

(5) *the morphism $0 \to Z$ is an element of S,*
(6) *the morphism $Z \to 0$ is an element of S,*
(7) *there exists a distinguished triangle (X, Y, Z, f, g, h) such that $f \in S$.*

Proof. The equivalence of (1), (2), and (3) is Homology, Lemma 05QF. If (2) holds, then $(Z'[-1], Z'[-1] \oplus Z, Z, (1,0), (0,1), 0)$ is a distinguished triangle (see Lemma 05QT) with "$0 \in S$". By rotating we conclude that (4) holds. If $(X, Y, Z \oplus Z', f, g, h)$ is a distinguished triangle with $f \in S$ then $Q(f)$ is an isomorphism hence $Q(Z \oplus Z') = 0$ hence $Q(Z) = 0$. Thus (1) – (4) are all equivalent.

Next, assume that S is saturated. Note that each of (5), (6), (7) implies one of the equivalent conditions (1) – (4). Suppose that $Q(Z) = 0$. Then $0 \to Z$ is a morphism of $\mathcal{D}$ which becomes an isomorphism in $S^{-1}\mathcal{D}$. According to Categories, Lemma 05Q9 the fact that S is saturated implies that $0 \to Z$ is in S. Hence (1) $\Rightarrow$ (5). Dually (1) $\Rightarrow$ (6). Finally, if $0 \to Z$ is in S, then the triangle $(0, Z, Z, 0, \mathrm{id}_Z, 0)$ is distinguished by TR1 and TR2 and is a triangle as in (7). $\square$